Quantification: 548_hyp_HP_z20c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	18
interior lacuna count	14
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 14)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2781.57	1453	2388.84	px^2
major_axis_length_px	lacuna shape	84.7868	73.0575	38.265	px
minor_axis_length_px	lacuna shape	42.3424	34.9392	19.9537	px
aspect_ratio	lacuna shape	2.1105	2.0264	0.6114	unitless
eccentricity	lacuna shape	0.8464	0.8698	0.0942	unitless
solidity	lacuna shape	0.7793	0.7846	0.0862	unitless
roots_count	headline	9.6429	9	5.3581	count
ring_length_r30_px	headline	311.929	298	139.803	px
ring_length_r60_px	reported	628.786	590.5	248.444	px
owned_length_px	ownership-dependent	869.718	668.949	559.803	px
edge_count	ownership-dependent	47.0714	39.5	29.6971	count
mean_edge_length_px	ownership-dependent	18.3349	17.7119	2.712	px
perimeter_px	normalised	302.359	247.028	176.653	px
ring_area_r30_px2	normalised	9380.07	8093.5	3556.74	px^2
ring_area_r60_px2	normalised	20606.6	18112.5	7135.73	px^2
in_frame_fraction_r30	normalised	0.9836	1	0.0417	unitless
in_frame_fraction_r60	normalised	0.9528	1	0.0845	unitless
ring_density_r30	normalised	0.0328728	0.0333611	0.00678406	px^-1
ring_density_r60	normalised	0.0306568	0.0335278	0.00602686	px^-1
roots_per_100px_perimeter	normalised	3.4346	3.2008	1.3109	per 100 px
ring_attached_length_r30_px	network v2	274.929	262.5	135.794	px
ring_attached_length_r60_px	network v2	462.429	425.5	233.384	px
ring_length_w_r30_px	network v2	354.971	337.628	158.29	px
ring_length_w_r60_px	network v2	719.819	672.42	287.386	px
sholl_crossings_r10	network v2	7.2857	7	3.0742	count
sholl_crossings_r20	network v2	7.7857	7.5	3.704	count
sholl_crossings_r30	network v2	7.8571	8	3.1097	count
loop_count	loops, report only	2.7143	2	2.9202	count
loop_edge_share	loops, report only	0.1589	0.1522	0.1301	share
ring_width_mean_r30_px	width	7.259	7.027	0.6035	px

Per field
----

measure	value	units
analysed_area_px2	1.00106e+06	px^2
total_skeleton_length_px	25545	px
canalicular_length_density_per_px	0.0255181	px^-1
junction_count	527	count
junction_density_per_px2	0.00052644	px^-2
lacuna_count	18	count

lacunae_per_px2	1.798e-05	px^-2
skeleton_component_count	429	count
mean_component_length_px	59.5455	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0290688	px^-1
field_density_without_flagged_per_px	0.024252	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	541	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0255181	px^-1
field_length_density_with_unattached_per_px	0.0260585	px^-1
loop_count	55	count
loop_edge_share	0.1388	share
edge_count	1463	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4729	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	21895	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1331	68.1738	43.3369	1.5731	0.772	0.5571	0.8707	22.9699	562.642	True
2	1345	60.6773	30.0142	2.0216	0.8691	0.8739	0.6556	34.9123	469.557	False
3	5304	115.818	61.0253	1.8979	0.8499	0.8497	-0.5279	76.0664	899.181	False
4	1100	56.8071	27.1347	2.0935	0.8785	0.8035	0.1617	61.0436	700.865	False
5	1020	65.1958	29.2967	2.2254	0.8933	0.5543	0.7245	258.288	921.093	False
6	2999	76.2507	53.7089	1.4197	0.7098	0.8844	-0.1093	288.357	25.6549	True
7	8181	176.425	74.8525	2.357	0.9055	0.7394	-0.9469	333.681	384.601	False
8	772	43.3165	23.8026	1.8198	0.8355	0.8863	1.0039	324.626	799.803	False
9	843	67.1384	19.7863	3.3932	0.9556	0.7938	-0.8751	303.813	882.996	False
10	3628	78.9766	63.2544	1.2486	0.5988	0.8761	0.6151	443.631	464.428	False
11	1526	56.8255	39.8642	1.4255	0.7126	0.7592	0.9788	480.14	672.478	False
12	2634	99.9739	44.742	2.2345	0.8943	0.697	1.1806	489.293	203.02	False
13	5796	118.655	71.2141	1.6662	0.7999	0.7445	0.2563	543.606	765.066	False
14	527	38.3444	21.2336	1.8058	0.8327	0.7784	0.8959	666.97	249.159	False
15	2266	83.3012	36.8142	2.2627	0.897	0.8389	0.7457	720.831	996.501	True
16	1380	84.7713	25.484	3.3264	0.9537	0.7908	-1.481	787.319	890.853	False
17	1983	87.7473	31.9283	2.7483	0.9315	0.8015	-1.3962	877.209	988.193	True
18	4886	124.09	61.0897	2.0313	0.8704	0.7632	0.9159	965.955	722.24	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	203	364	593.186	19	31.2203	10.1383

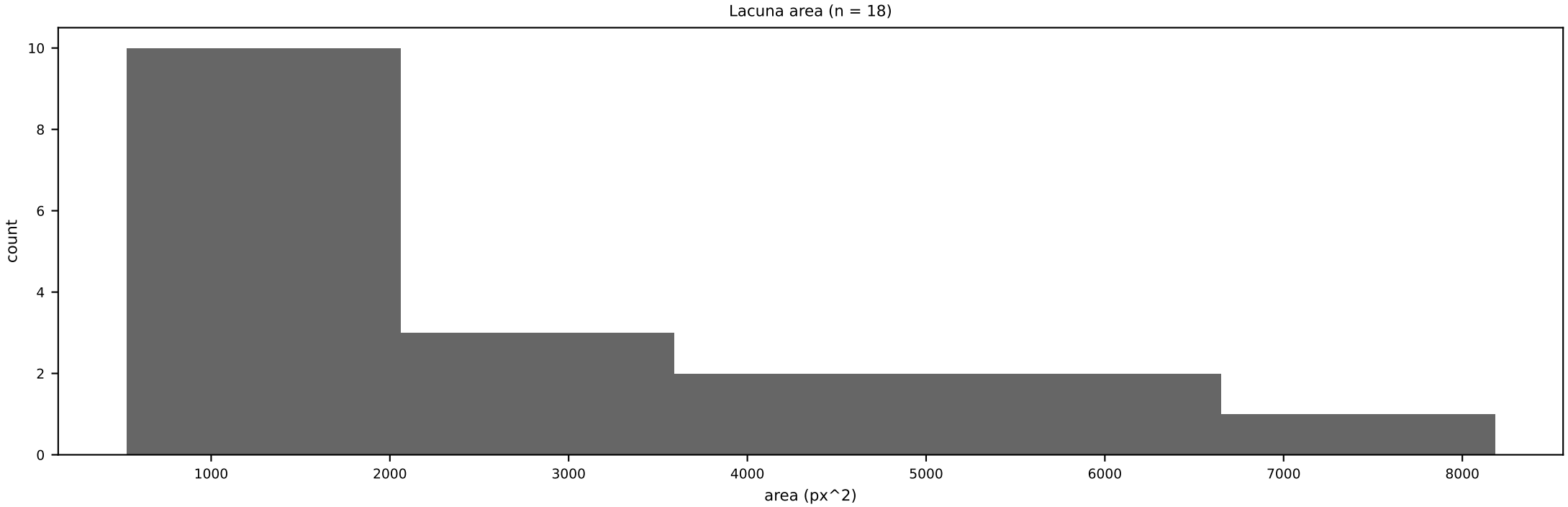
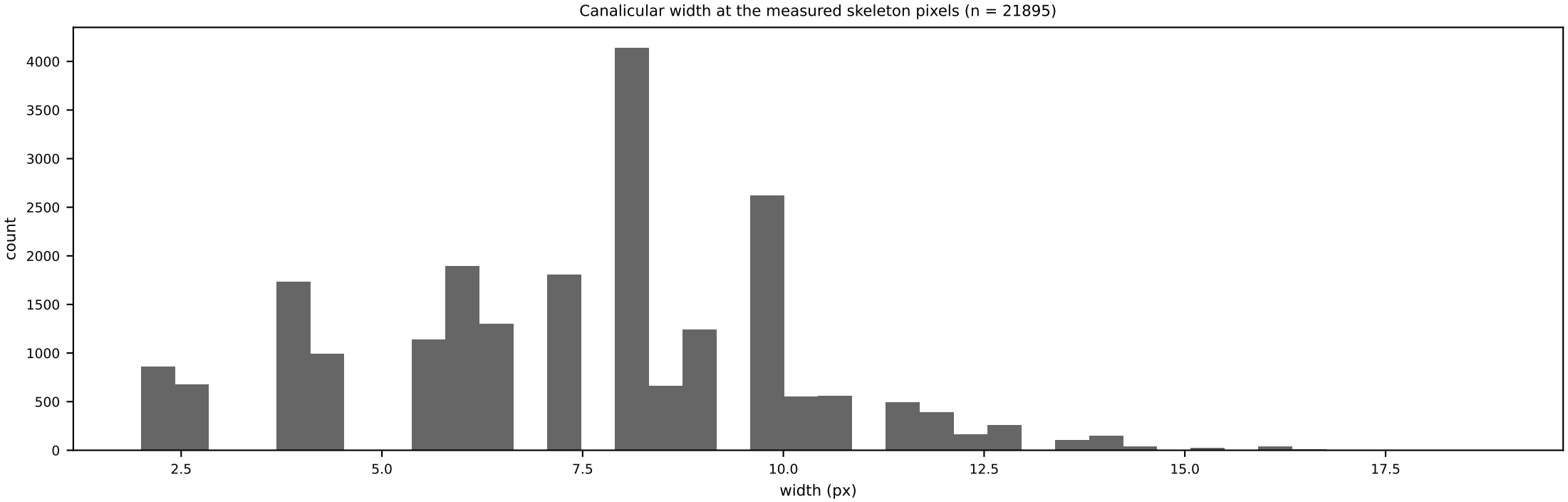
2	4	237	451	414.576	17	24.3868	6.6843
3	9	366	781	1267	62	20.4355	8.385
4	9	316	578	697.571	39	17.8864	6.9559
5	12	274	600	323.907	20	16.1953	7.3181
6	5	215	430	944.021	53	17.8117	7.6771
7	24	658	1200	1938.38	101	19.1919	6.7801
8	3	181	249	350.747	20	17.5374	6.9034
9	7	151	354	319.563	19	16.8191	6.4736
10	12	355	734	638.316	40	15.9579	6.7865
11	13	294	602	1244.67	62	20.0754	7.552
12	11	423	910	640.328	34	18.8332	7.5163
13	13	450	837	1681.22	96	17.5127	8.2997
14	6	100	405	850.394	40	21.2599	7.0981
15	5	262	440	165.225	9	18.3584	7.2195
16	4	260	521	1520.82	87	17.4807	7.947
17	7	252	529	571.872	26	21.9951	8.3853
18	8	302	581	288.551	22	13.116	6.926

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	299.12	6629	11618	0.7226	0.6609	0.030623	0.0313307	2.0059
2	185.238	6495	12887	0.8654	0.7462	0.0364896	0.0349965	2.1594
3	355.505	12183	26255	0.996	0.8914	0.0300419	0.0297467	2.5316
4	180.894	7279	16835	1	0.8909	0.0434126	0.0343332	4.9753
5	247.806	7577	17555	1	1	0.0361621	0.0341783	4.8425
6	300.635	6752	15901	0.711	0.6516	0.0318424	0.0270423	1.6631
7	732.927	17463	35109	1	1	0.0376797	0.0341793	3.2745
8	129.095	5795	14290	1	1	0.0312338	0.0174248	2.3239
9	153.238	5367	9246	1	1	0.0281349	0.0382868	4.5681
10	300.433	10348	22315	1	1	0.0343062	0.0328927	3.9942
11	246.25	7528	16500	1	1	0.0390542	0.0364848	5.2792
12	351.747	10779	26637	1	1	0.039243	0.034163	3.1272
13	533.985	13882	29213	1	1	0.0324161	0.0286516	2.4345
14	118.953	5937	17484	1	1	0.0168435	0.023164	5.044
15	256.108	6631	13116	0.7153	0.654	0.0395114	0.0335468	1.9523
16	213.137	8610	18670	1	1	0.0301974	0.0279057	1.8767
17	256.551	6858	14534	0.748	0.6875	0.0367454	0.0363974	2.7285
18	483.813	12078	25496	0.909	0.8104	0.0250041	0.0227879	1.6535

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	170	268	230.894	406.149	8	7	6
2	232	269	269.622	504.09	6	5	4
3	311	711	412.019	890.495	8	13	10
4	273	427	357.492	648.316	7	9	7
5	252	314	308.764	687.955	7	7	8
6	149	225	248.936	499.215	6	7	4
7	632	1022	753.097	1397.51	16	15	13
8	151	185	211.137	284.664	6	5	5
9	108	270	178.581	421.789	3	2	3
10	280	433	392.321	819.34	7	9	11
11	294	542	338.907	694.784	6	8	6
12	302	482	474.718	1040.8	9	9	11
13	439	776	517.76	960.649	10	11	10
14	87	356	115.811	462.061	4	2	4

15	140	232	290.137	492.647	5	5	10
16	249	424	303.007	608.127	6	7	10
17	218	418	286.693	610.571	6	6	9
18	239	263	336.35	656.884	7	7	8

548_hyp_HP_z20c1-2: distributions, pre-validation, px